Golomb Rulers - 2

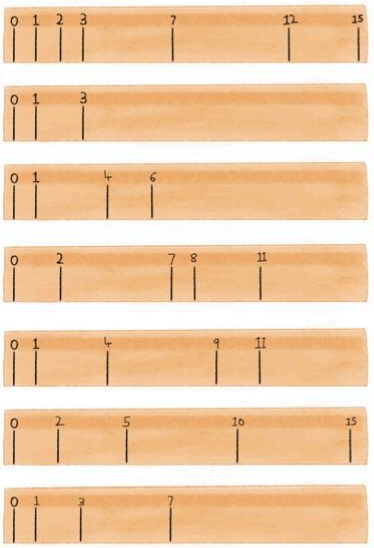
Grade 8: Number Play

Introduction

An interesting function of computers is to make them faster by figuring out ways to perform a task with the minimum amount of information. We need to develop efficient, fast methods to solve problems, but we also need to optimise the amount of information required to complete a task. The less information needed, the less computer memory required, the more efficient the overall process is.

In this activity, we will explore the idea of minimising the amount of information needed to perform a task by looking at an interesting type of ruler known as a Golomb ruler. The speciality of these rulers is that even though they do not have all the markings in a typical ruler, they can still measure all the integer lengths uniquely. The task the rulers perform is measuring integer lengths; a smaller number of markings corresponds to less information needed to do the task.





Activity	Time	Description
Launch	5 min	The teacher shows students the rulers and introduces the idea of Golomb rulers.
Problem-solving and algorithm	25 mins	Student's Tryout the activity with each other. Student Worksheet: https://tinyurl.com/goulombruler1worksheet

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